

A Note on Relevance

Diego Feinmann

Institute of Computer Science Polish Academy of Sciences

Abstract

In this paper, I present novel data that challenge the dominant approach to modelling linguistic relevance, according to which a sentence is relevant if and only if it is informative to the Question Under Discussion (QUD). In particular, I show that a sentence can fail to be informative to the QUD and, despite this, still be perceived as relevant. In the final section, I outline the theoretical implications of this finding.*

1 Introduction

In Gricean and Neo-Gricean approaches to pragmatics, utterances are assumed to be mediated by a series of conversational maxims. One such maxim is the Maxim of Relation (Grice's original terminology), also known as the Maxim of Relevance, which instructs speakers to make contributions that are related or relevant, in Grice (1975: 47) own words, 'to the immediate needs [...] of the transaction'. Grice left two issues open, however: (i) How should 'the immediate needs of the transaction' be characterised? (ii) What does it take for a linguistic contribution to count as relevant to 'the immediate needs of the transaction'?

Theoretical work carried out during the 80s and 90s—in particular, Groenendijk and Stokhof's (1984) ground-breaking work on questions and the seminal Roberts ([1996] 2012)—filled these two gaps in a rather compelling manner:

- (i) Groenendijk and Stokhof (1984) identified 'the immediate needs of the transaction' with 'the topic of the conversation,' and the latter with (the meaning of) the question that interlocutors are concerned with at a given point in discourse¹—formally represented in their framework as a partition of a set of worlds. Since Roberts ([1996] 2012), this partition has been known as the *Question Under Discussion* (QUD) and the set of worlds over which it is defined has been identified with the context set, i.e. the set of worlds compatible with the common ground (Stalnaker 1978).

* I am grateful to Justin Bledin, Keny Chatain, Paul Egré, Dan Hoek, Adam Przepiórkowski, Sonia Ramotowska, Jacopo Romoli, Daniel Rothschild, Benjamin Spector, and Chris Kennedy, the editor, for their valuable comments and suggestions. Any remaining errors are my own.

¹ Groenendijk and Stokhof (1984: 6): 'A notoriously difficult, but quite essential maxim proposed by Grice, is the Maxim of Relation. Relevance, it seems, is essentially tied to what a conversation is about, to what the topic of a conversation is. And a topic of conversation may very well be thought of as a (set of) question(s). This is obvious for discourses which consist of explicit question-answer sequences, but seems to hold also for types of conversation that are not explicitly concerned with information exchange. Even if in some discourse, no question is explicitly raised, it still plays an important role at the background, viz. as the topic that makes the discourse a coherent whole, rather than a random sequence of assertions. *The topic, i.e. an explicitly or implicitly raised question, is what defines the relevance of the assertions in a discourse for each other* [italics mine].'

- (ii) Groenendijk and Stokhof (1984) conceptually linked the notion of relevance to that of question answering: a sentence is relevant with respect to a QUD Q if and only if it resolves, at least partially, Q . That is, to count as relevant, a sentence does not need to determine in which cell of Q the actual world is, but it must at least rule out one cell as a possible location.

The precise formulation of the above is captured in (D-I):

(D-I) **Standard QUD Relevance**

Let ϕ be a declarative sentence, c the context set (the set of worlds compatible with the common ground), and Q a QUD (a partition of c).

- (a) ϕ is relevant to Q iff ϕ gives a partial answer to Q ;
(b) ϕ gives a partial answer to Q iff $\llbracket \phi \rrbracket \cap c \neq \emptyset$ and there exists a cell $q \in Q$ s.t. $q \cap \llbracket \phi \rrbracket = \emptyset$.^{2,3}

(In words: ϕ gives a partial answer to Q iff ϕ is not a contextual contradiction, and its truth-conditions fail to intersect, and hence they eliminate, at least one cell of Q).

I shall refer to (D-I)—which is nothing else but Groenendijk and Stokhof’s (1984) definition of relevance⁴ with some minor updates⁵—as *Standard QUD Relevance* (*SQR* for short).

To see how *SQR* works, consider the following example:

- (1) A: Is Mary coming to the party tonight?
B₁: She is (coming). [Relevant ✓]
B₂: Jeff Bezos hates Bill Gates. [Relevant ✗]

A’s question induces a QUD consisting of just two cells—the YES-cell (the set of worlds in which Mary is coming to the party) and the NO-cell (the set of worlds in which she isn’t coming). (1)B₁ gives a partial answer to this QUD and is therefore relevant to it: (1)B₁ is not a contextual contradiction (if it was common ground that Mary isn’t coming, A wouldn’t be asking such a question) and eliminates the

² $\llbracket \phi \rrbracket$ is a shorthand for $\{w \mid \llbracket \phi \rrbracket^w = 1\}$, i.e. ϕ ’s truth-conditions (the set of worlds in which ϕ is true).

³ The provision that a partial answer cannot be a (contextual) contradiction is present in all Groenendijk and Stokhof’s definitions of partial answerhood I’m aware of—semantic ones, where question denotations are partitions of logical space, and pragmatic ones, where they are restricted to an information set (cf. Groenendijk and Stokhof 1984; 1990). If one disagrees with this provision, i.e. if one thinks that contradictions should count as relevant (cf. Bar-Lev 2024), then one can just drop ‘ $\llbracket \phi \rrbracket \cap c \neq \emptyset$ and’ from (1). The empirical issues that will be discussed here are completely orthogonal to this debate.

⁴ See Groenendijk and Stokhof’s (1984) Definition (38) on page 242, which is based on the notion of *giving a partial (pragmatic) answer*. This notion is introduced in prose on page 233 and later given an explicit definition in (28) on page 353. (Note: there is a typographical error in (28): ‘=’ should be ‘ $\subseteq$ ’, as clarified in Groenendijk and Stokhof (1990: 22)—see Definition 21.2.)

⁵ ‘Question’ talk is replaced by ‘QUD’ talk, the notation is simplified, and the partial answerhood condition is expressed in a more concise manner. Moreover, QUDs are explicitly characterised as partitions of c (the Stalnakerian context set), as in Roberts ([1996] 2012) and most subsequent literature. (In Groenendijk and Stokhof’s (1984) definition, QUDs are also characterised as partitions of an information set (a subset of logical space), though not much is said about the nature of this set beyond that it should be thought of as a ‘representation of the information of the questioner’ (Groenendijk and Stokhof 1984: 351).

NO-cell, i.e. if (1) B_1 is the case, then it must be the case that the actual world lives somewhere in the YES-cell. (1) B_2 , by contrast, is not relevant to the QUD: the proposition it expresses is compatible with Mary coming to the party as well as with her not coming, and thus it eliminates no QUD cell.

SQR, albeit elegant, is rather limited—as observed, for example, in Simons et al. (2010: 316-7), it fails to diagnose relevance in cases in which the sentence merely raises the probability of the world being in some cell or other but doesn’t rule out any given cell (see also Roberts 2012). Consider (2), for example.

- (2) A: Is Mary coming to the party tonight?
 B_1 : She has an exam tomorrow morning. [Relevant ✓]
 B_2 : Jeff Bezos hates Bill Gates. [Relevant ✗]

It seems clear that, unlike (2) B_2 , (2) B_1 is a relevant thing to say as a response to A’s question. According to *SQR*, however, both (2) B_1 and (2) B_2 are irrelevant: (2) B_1 , just like (2) B_2 , is compatible with both QUD cells and hence rules out none—indeed, it’s possible for (2) B_1 to be true and for Mary not to attend the party; and it’s possible for (2) B_1 to be true and for Mary to attend the party (she might not care about the exam, etc.). In other words, by uttering (2) B_1 , the speaker communicates that Mary is unlikely to attend the party, but not that she will certainly not attend.⁶

To make sense of the contrast in (2), a probabilistic definition of relevance can be given—one that, at a certain level of abstraction, is conceptually analogous to *SQR*. Indeed, at a certain level of abstraction, *SQR* just tells us that, for a sentence φ to be relevant to a QUD, it must, in addition to not being a contextual contradiction, be *informative* relative to the QUD. In a non-probabilistic framework, ‘being informative relative to the QUD’ just means eliminating at least one QUD cell (downgrading at least one cell from possible to impossible); in a probabilistic framework, ‘being informative relative to the QUD’ amounts, at a minimum, to having a non-trivial effect on the probabilities of the QUD cells, i.e. to shifting the probabilities assigned to those cells. Thus, in a probabilistic framework, relevance can be defined as follows:

⁶ This fact is exposed in full light in the following minimal pair:

- (i) A: Is Mary coming to the party tonight?
 B: She has an exam tomorrow morning. (✓) So, she probably won’t.
 (ii) A: Is Mary coming to the party tonight?
 B: She is not. (#) So, she probably won’t.

(D-II) *Probabilistic QUD Relevance*

Let P be the common prior, a probability distribution over Ω (the set of all possible worlds); let Q be a QUD (now defined as a partition of Ω ⁷); and let φ be a declarative sentence (see notational convention⁸).

φ is relevant to Q under P iff $P(\varphi) > 0$, and there exists a cell $q \in Q$ such that $P(q) \neq P(q \mid \varphi)$.

(Equivalently, in words: φ is relevant to Q under P iff the truth of φ is not ruled out by P , and conditioning P on the truth of φ shifts the probabilities assigned to the cells of Q .⁹)

To the best of my knowledge, the definition in (D-II), which I'll refer to as *Probabilistic QUD Relevance* (*PQR* for short), was first sketched in Büring (2003). It should be noted, however, that the idea of defining relevance in probabilistic terms is an old one. Carnap (1950), for example, characterised relevance as follows (with some minor notational modifications that preserve the original idea): a sentence φ is relevant to a hypothesis h (a proposition) under a distribution P iff $P(\varphi) > 0$ and $P(h) \neq P(h \mid \varphi)$. The only significant difference between this definition and *PQR* is that, in the former, the relevance of a sentence is defined with respect to a proposition and a probability distribution, whereas in *PQR* it is defined with respect to a partition of Ω (a set of propositions) and a distribution. The mechanics are identical.

PQR, which is weaker than *SQR*, predicts the contrast in (2), provided that the prior distribution is such that having an exam is a likely reason for Mary not to attend the party whereas Bezos hating Gates has no bearing on Mary's decision to attend. Indeed, given such a prior, (2) B_1 is bound to induce a shift of probability mass from the YES-cell to the NO-cell, whereas (2) B_2 is expected to have no effect on the cell probabilities.

It is worth noting that, although *PQR* can account for the contrast in (2), it is not suited to explain, for example, why (2) B_1 , though intuitively relevant, is not as relevant as, for example, (1) B_1 . In other words, *PQR*, just like *SQR*, tells us whether a declarative sentence is relevant or not, but it doesn't tell us *how* relevant it is. To answer the latter question, a general method for assigning a relevance score to a sentence/proposition is required. Such methods are readily available in the literature—see, for example, Russell (2012) for a scoring system based on Carnap's proposed measure of relevance, or Feinmann

⁷ In Bayesian models of communication, the shared background state is modeled by a probability distribution P over Ω (the common prior), rather than by a subset of Ω (the context set), and QUDs can be viewed as partitions of Ω . Worlds not compatible with the shared background state are not excluded from the QUD—as in the non-probabilistic framework—but are simply assigned probability 0 under P .

⁸ **Notational convention.** Throughout this paper, lowercase Greek letters ($\varphi, \psi, \dots$) name sentences. To avoid clutter, when such a letter appears inside a probability expression (as an argument of P or as a conditioning event), it stands for the truth-conditions of that sentence; e.g. $P(\varphi)$ abbreviates $P(\llbracket \varphi \rrbracket)$ and $P(q \mid \varphi)$ abbreviates $P(q \mid \llbracket \varphi \rrbracket)$.

⁹ If there exists a cell $q \in Q$ such that $P(q) \neq P(q \mid \varphi)$, then one of two things must be true: either $P(q) < P(q \mid \varphi)$, in which case there must be at least one other cell $q' \in Q$ such that $P(q') > P(q' \mid \varphi)$; or $P(q) > P(q \mid \varphi)$, in which case there must be at least one other cell $q' \in Q$ such that $P(q') < P(q' \mid \varphi)$. This follows because Q partitions Ω , so the total probability across its cells is 1 both before and after conditioning on the truth of φ .

(2020) for a relevance scoring system based on Kullback–Leibler (KL) divergence. Here I won’t be concerned with the issue of gradience in relevance judgments, so I won’t discuss any of these methods.

As already mentioned, *SQR* and *PQR* can be conceptualised, at least at a certain level of abstraction, as different implementations of a single core idea, which can be summarised as follows:

Informativity-based QUD Relevance

- (i) The relevance of a sentence is determined relative to a QUD and an epistemic state.
- (ii) A sentence is relevant iff, given what is known/believed, it is *informative* to the QUD, i.e. the proposition it expresses has a non-trivial effect on the QUD cells.

In *SQR*, the epistemic state, at least since Roberts ([1996] 2012), is identified with the Stalnakerian context set (the set of worlds compatible with the common ground); in the *PQR*, in turn, the epistemic state is identified with the ‘common prior’ (a probability distribution).¹⁰ In *SQR*, ‘being informative’ amounts to ruling out cells, i.e. *a priori* possible ways of resolving the question; in *PQR*, ‘being informative’ amounts to having a non-trivial effect on the probabilities of the QUD cells.

In the next section, I present some data that call into question this conception of relevance.

2 Not all uninformative sentences are irrelevant

As discussed above, *Informativity-based QUD Relevance* (henceforth, *Info-QUD-R*), whether formalised as *SQR* or *PQR*, entails that the uttered sentence is doomed to irrelevance unless it is *informative* to the QUD. With this in mind, consider the following contrast:

(3) A: Is Mary coming to the party tonight?

- B₁: Mary’s father told me she’s coming, but Mary’s mother said the opposite. [Relevant ✓]
B₂: Jeff Bezos hates Bill Gates. [Relevant ✗]

(3)B₁ seems like a relevant contribution given the question that has been raised: what B₁ is saying is clearly *related to* A’s question, unlike the response in (3)B₂, which is clearly *unrelated*. From the *SQR* perspective, this contrast is unexpected: both sentences intersect every cell of the QUD induced by (3)A, and thus both sentences are predicted to be irrelevant. The contrast in (3) is, therefore, just as problematic for *SQR* as the one in (2). The latter, however, does not pose a fundamental threat to *Info-QUD-R*—as discussed, *PQR*, the probabilistic implementation of *Info-QUD-R*, can be invoked to make sense of (2). (3), however, is different—it poses as much of a challenge to *SQR* as it does to *PQR*.

¹⁰ See Morris (1995) for a discussion on the common prior assumption.

The issue is this: the first conjunct of (3) B_1 raises the probability of the QUD’s YES-cell and lowers that of the NO-cell, whereas the second conjunct does precisely the opposite—it lowers the probability of the YES-cell and raises the probability of the NO-cell. Assuming there’s no reason to believe that one parent is more likely than the other to provide truthful information—a simplifying assumption that, for present purposes, amounts to assuming that each conjunct of (3) B_1 shifts the odds in opposite directions but with the same strength—learning that (3) B_1 is true is bound to have no net effect on the QUD cells: in such a configuration, the two shifts cancel each other out. *PQR*, therefore, is unable to make sense of the contrast between (3) B_1 and (3) B_2 : under the stated assumptions, neither (3) B_1 nor (3) B_2 shifts the probabilities of the QUD cells.

The empirical point made by (3), it should be noted, has a historical precedent. Keynes (1921), to my knowledge, was the first to introduce the conception of relevance that lies at the heart of *PQR* and is often attributed to Carnap (1950), i.e. a statement ϕ is relevant to a hypothesis h under a distribution P iff $P(\phi) > 0$ and $P(h) \neq P(h \mid \phi)$. Keynes, however, was unsatisfied with this notion; on his view, a statement of the form ‘ $\phi \wedge \psi$ ’ should count as relevant to a given hypothesis h even when $P(h) = P(h \mid \phi \wedge \psi)$, provided that $P(h) \neq P(h \mid \phi)$ and $P(h) \neq P(h \mid \psi)$.¹¹ This (conceptual) intuition—aligned with the judgments reported in (3)—led Keynes (1921) to dismiss his own characterisation of relevance, which Carnap (1950) later revived.¹²

Now, if we look at (3) B_1 with a linguistic eye, the fact that (3) B_1 is intuitively relevant could be interpreted, not as fundamental problem for *PQR* as a relevance-assessing mechanism, but rather as an indication that this mechanism should be computed in stages, as interpretation proceeds, rather than on the global meaning of the uttered sentence. Indeed, what appears to be happening in (3) B_1 is this: first, one hears ‘Mary’s father told me she’s coming’, which, according to *PQR*, is relevant (it raises the probability of the QUD’s YES-cell); then, one learns that ‘Mary’s mother said the opposite’, which, according to *PQR*, is also relevant (it raises the probability of the QUD’s NO-cell). Thus, insofar as one holds to the view that *PQR* can target different coordinate clauses as interpretation proceeds, one might not be too worried about (3) B_1 —though a problem for *PQR* as it stands, it might be possible to provide a dynamic/incremental rendition of *PQR* that circumvents this problem.

The issue, however, appears to be deeper than what (3) suggests. Consider, for example, (4):

¹¹ In Keynes’ (1921: 72) own words: ‘If we are able to treat “weight” and “relevance” as correlative terms, we must regard evidence as relevant, part of which is favourable and part unfavourable, even if, taken as a whole, it leaves the probability unchanged.’

¹² Anecdotally Keynes did formulate an alternative account of relevance, which Carnap proved untenable three decades later by showing it leads to trivialisation (see Carnap 1950: 419-20).

(4) A: Is Mary coming to the party tonight?

B₁: Mary's parents gave me conflicting reports.

[Relevant ✓]

B₂: Jeff Bezos hates Bill Gates.

[Relevant X]

(4)B₁ has no effect on the QUD cells, nor does (4)B₂ (at least under reasonable assumptions about the prior distribution). Despite this, there's a clear contrast between the two: (4)B₁, just like (3)B₁, is intuitively relevant, while (4)B₂ is not. In contrast to (3)B₁, however, there appears to be no principled way to decompose (4)B₁ into two or more clauses such that (i) updating these clauses in sequence produces the same outcome as updating the full sentence at once, and (ii) each step in the sequence has a non-trivial effect on the probabilities of the QUD cells. Since the problem posed by (4)B₁ is, as far as I can tell, the same problem as that posed by (3)B₁, I don't see how a refinement of *PQR* along the lines sketched above could offer a way forward.¹³

3 *Info-QUD-R*: final assessment

For Grice, if a linguistic contribution was related to 'the immediate needs of the transaction,' then it was relevant to it, and vice versa—let's remember that he equates the Maxim of Relation with the command 'Be relevant'. In the early 1980s, Groenendijk and Stokhof came up with a very influential idea, that of conceptualising Grice's Maxim of Relation along the following lines: a linguistic contribution is relevant/related to the QUD (the topic of the conversation) if and only if it is informative to the QUD (in their framework, 'being informative' boils down to 'giving a partial answer'; see §1). Now, if one takes our observations at face value, it's clear that the notion of being relevant in the sense of being *related* cannot be assimilated to the notion of being relevant in the sense of being *informative*: a sentence may not be informative to the QUD (it may not have any detectable effect on the QUD cells) but may still be related to the QUD, as the contrast between (3)B₁ and (3)B₂ and that between (4)B₁ and (4)B₂ indicate.¹⁴

This point can be amplified by considering possible continuations that an uninformative yet relevant/related answer might allow *vis-à-vis* an answer that is both uninformative and irrelevant/unrelated. In particular, the former kind of answer cannot be challenged with the follow-up 'And what does that have to do with anything?' whereas the latter can, as shown below.

(5) A: Is Mary coming to the party?

B: Mary's parents gave me conflicting reports.

[uninformative & related to the QUD]

A: # And what does that have to do with anything?

¹³ Thanks to Daniel Rothschild for the example in (4)B₁ and helpful discussions on this issue.

¹⁴ Thanks to Paul Egré and Adam Przepiórkowski for fruitful exchanges on this matter.

- (6) A: Is Mary coming to the party?
B: Jeff Bezos hates Bill Gates. [uninformative & unrelated to the QUD]
A: ✓ And what does that have to do with anything?

In the light of these data, I don't think that *Info-QUD-R* can be said to successfully model Grice's Maxim of Relation: it fails to explain why (5)B is perceived as being related to the question raised, whereas (6)B is not (or, to restate this in terms of the diagnostic introduced above: it fails to account for the divergent behaviour of (5)B and (6)B under the *And what does that have to do with anything?* test.) I would like to stress that I have no qualms with *Info-QUD-R* as such—it is a perfectly natural notion, one that has empirical reflexes of its own.¹⁵ My point is simply that *Info-QUD-R* rules out as irrelevant contributions that, albeit uninformative, do not give rise to a judgment of irrelevance.

As I see it, the data discussed here can be interpreted in at least two ways:

- (i) *Info-QUD-R* is not the right approach to modelling linguistic relevance.
- (ii) *Info-QUD-R* is an important ingredient of 'the relevance algorithm' but is not the only ingredient. On this view, *Info-QUD-R* does play a role in the production of our judgments of relevance *in conjunction with* other considerations, which are yet to be specified.

At this point, I have no particular reason to favour one interpretation over the other. That said, in these final pages, I would like to consider four perspectives that fall under (ii), each of which, at least on first inspection, appears to have some degree of plausibility. In doing so, I shall take *PQR*—the probabilistic implementation of *Info-QUD-R*—as the framework of reference, as it offers broader explanatory scope than *SQR*, its non-probabilistic counterpart.

3.1 QUD accommodation

Perhaps all that is needed to make sense of our data is to supplement *PQR* with an account of QUD accommodation. From this perspective, (5)B, repeated below as (7)B, would be perceived as relevant because it would be interpreted not as an answer to the QUD induced by the overt question but as an answer to an accommodated QUD, one relative to which *PQR* diagnoses the sentence as relevant. Conversely, (6)B, repeated below as (8)B, would be perceived as irrelevant because it would not be possible to accommodate a QUD relative to which *PQR* diagnoses the sentence as relevant.

¹⁵ For example, it would be perfectly natural to respond to an uninformative yet relevant answer like (5)B with 'Well, that's not very helpful...'. By contrast, it would be odd to respond that way to an informative answer—(1)B₁ and (2)B₁, quite clearly, do not admit such a rejoinder.

(7) A: Is Mary coming to the party tonight?
B: Mary's parents gave me conflicting reports. [Relevant ✓]

(8) A: Is Mary coming to the party tonight?
B: Jeff Bezos hates Bill Gates. [Relevant X]

It is not clear to me how to spell out a constraint on QUD accommodation that accomplishes this. (7)B could be taken to address the QUD in (9): (7)B is clearly informative relative to such a QUD, as it eliminates those cells in which Mary's parents agree on their guesses.

(9) What are Mary's parents' best guesses as to whether Mary is attending the party?
{ {mother and father think she's coming}, {mother and father think she's not coming}, {mother thinks she's coming, and father thinks she's not}, {father thinks she is coming, and mother thinks she is not}, etc.¹⁶}

However, by the same token, (8)B could be taken to address the QUD in (10), which would lead to a bad prediction: it would predict (8)B to be perceived as relevant, contrary to fact.

(10) Does Jeff Bezos hate Bill Gates?
{ {jeff bezos hates bill gates}, {jeff bezos doesn't hate bill gates} }

To address this problem, one could put forth the following constraint on QUD accommodation:

(D-III) Let φ be the uttered sentence, P the common prior, Q the QUD induced by the overt question, and Q^* any other QUD.

If φ is diagnosed as irrelevant by PQR —that is, if $P(q) = P(q \mid \varphi)$ for every $q \in Q$ —one may reject Q and accommodate Q^* only if the following two conditions are met:

- (i) Conditioning P on the truth of φ shifts the probabilities of the cells of Q^* .
- (ii) There exists a cell $q \in Q^*$ such that conditioning P on q shifts the probabilities of the cells of Q .

(D-III) blocks the QUD in (10) from being accommodated in (8): there is no cell in (10) such that conditioning P on it shifts the probabilities of the (8)A-QUD cells (whether Jeff Bezos hates Bill Gates, or whether he doesn't, is orthogonal to the question of whether Mary is coming to the party).

¹⁶ To avoid clutter, I'm not listing cells in which at least one of the parents is maximally uncertain about Mary's plans.

(D-III), by contrast, doesn't interfere with the possibility of accommodating (9) in (7). Indeed, in this case, there's a cell in (9) (e.g. {mother and father think she's not coming}) such that conditioning P on it shifts the probabilities of the (7)A-QUD cells.

The issue, however, is this: there are in principle many QUDs that could be accommodated to 'rescue' (8)B. The constraint in (D-III) does prevent the QUD in (10) from being accommodated but, for example, is unable to filter out the QUD in (11).

(11) { {jeff bezos hates bill gates and mary has an exam tomorrow morning}, {jeff bezos hates bill gates and mary doesn't have an exam tomorrow morning}, {jeff bezos doesn't hate bill gates and mary has an exam tomorrow morning}, {jeff bezos doesn't hate bill gates and mary doesn't have an exam tomorrow morning} }

Indeed, conditioning P on the truth of (8)B shifts the probabilities of the cells of (11)—so condition (D-III)-i is met; furthermore, there is a cell in (11) (e.g. {jeff bezos hates bill gates and mary has an exam tomorrow morning}) such that conditioning P on it shifts the probabilities assigned to the cells of the QUD induced by (7)A (the overt question)—so condition (D-III)-ii is also met.

(D-III) is therefore too weak. A stronger constraint on QUD accommodation would be needed—one that prevents QUD accommodation from rescuing (8)B while allowing this mechanism to rescue (7)B. It is not clear to me whether such a constraint can be formulated—this is something I must leave for future research to determine.

3.2 Strategy of inquiry

Roberts ([1996] 2012) introduces the notion of *strategy of inquiry* (cf. Bratman 1987; Thomason 1990). Roughly speaking, we can think of a strategy of inquiry for answering a QUD Q as a set of sub-questions that, if answered, may resolve Q , or at least shift the probabilities of the cells of Q . This gives rise to a natural idea: even if it is diagnosed as irrelevant by PQR , a response may still be perceived as relevant, provided it suggests a strategy of inquiry—one that, if pursued, may uncover previously unknown facts that could, in turn, make the uttered sentence relevant by PQR standards.¹⁷

Let's again consider the exchange in (12):

(12) A: Is Mary coming to the party tonight?
B: Mary's father told me she's coming, but Mary's mother said the opposite.

¹⁷ I am grateful to an anonymous reviewer for suggesting this way of framing the point.

Upon hearing (12)B, one may be inclined to pursue the following strategy of inquiry: let's find out (i) at what time Mary last spoke with her mother, and (ii) at what time she last spoke with her father. Such a strategy could prove fruitful: one might find out that (i)'s answer is 3 p.m. on June 1st 2025 and (ii)'s answer is 8 p.m. on June 2nd 2025. Then (12)B, considered alongside these findings, would be informative and hence relevant in the *PQR* sense: it would now be apparent that the father, having spoken to Mary more recently, is likely to be a more reliable informant than the mother.

I should note that the strategy of inquiry just sketched would fail to rescue the non-conjunctive version of (12), repeated in (13) below:

- (13) A: Is Mary coming to the party tonight?
B: Her parents gave me conflicting reports.

Indeed, since in this case one doesn't know which parent said what, learning that one is likely to be better informed than the other doesn't break the tie. This is not necessary a problem for the strategy-of-inquiry view though—one might be able to implement a different strategy in such a case.

The problem, however, lies with those cases that are perceived as irrelevant—consider, once again, the exchange in (14).

- (14) A: Is Mary coming to the party tonight?
B: Jeff Bezos hates Bill Gates.

On the strategy-of-inquiry view, to account for the fact that (14)B gives rise to a judgment of irrelevance, it's not enough to say 'because it doesn't count as relevant according to *PQR*' (that's also true of (12)B and (13)B). Rather, one must say: 'because it doesn't count as relevant according to *PQR*, and moreover, no *PQR*-rescuing strategy of inquiry can be identified'. The question then arises: is this actually true?

I can at least think of one possible strategy: let's find out (i) whether Mary is married to Jeff Bezos, and (ii) whether the party in question is being organised by Bill Gates. It's not clear whether this strategy will ultimately succeed; however, if Mary turns out to be married to Bezos and Gates turns out to be the party's organizer, then (14)B—once combined with these findings—will, under a reasonable prior, have an effect on the probabilities of the QUD cells.

It might be argued that the strategy of inquiry that I've just outlined is too far-fetched, and that interpreters are therefore unlikely to come up with it 'on the spot'. But if that's the case, then a formal account of *strategy-of-inquiry plausibility* is needed, one that tells us which strategies can be carried out and which

cannot. On its own, the strategy-of-inquiry view does not explain the data: it offers a possible direction worth exploring but not a conclusive solution to the empirical issues raised here.

3.3 Relevance and ignorance

It is hardly news that one can always respond to a given question ‘ φ ?’ by saying ‘I don’t know (φ)’. This might be seen as an argument against *PQR*: an avowal of ignorance, unless a rather unnatural prior is assumed, is not expected to affect the probabilities of the QUD cells. But there’s an alternative take on this data point: ‘I don’t know (φ)’ is neither relevant nor irrelevant—rather, it is a way to *opt out* of answering ‘ φ ?’ Let’s assume that the latter perspective is the right one. Then it becomes tempting to claim that (15)B is just like ‘I don’t know’: by uttering this sentence, the speaker conveys that she doesn’t know the answer to (15)A and is therefore taken to have passed on the question rather than incurred a relevance violation.

- (15) A: Is Mary coming to the party tonight?
B: Mary’s parents gave me conflicting reports.
 $\rightsquigarrow$ B doesn’t know the answer to A’s question.

This perspective raises two issues, a theoretical and an empirical one. Let’s begin with the theoretical issue. (15)B communicates ignorance, yes, but it does so via *an implicature*, which is arguably mediated by Grice’s Maxim of Quantity: the speaker said something that doesn’t resolve the question; if s/he was in a position to resolve the question, s/he would have done so (because speakers are expected to make their contributions, in Grice’s (1975:45) own words, ‘as informative as is required (for the purposes of the exchange)’; therefore, the speaker must not know the answer to the question (i.e. the speaker must not know whether Mary is coming to the party or not).

If this implicature, often dubbed ‘ignorance inference’, is what saves (15)B from being perceived as irrelevant, then we are forced to address the following question: why isn’t ‘Jeff Bezos hates Bill Gates’ saved by the same mechanism? Note that, formally, ‘Jeff Bezos hates Bill Gates’ and ‘Mary’s parents gave me conflicting reports’ stand in the same relation to the QUD induced by ‘Is Mary coming to the party tonight?’: the truth of each of these sentences is compatible with the two QUD cells. The inference that the speaker is ignorant about whether Mary is coming to the party is therefore expected in both cases.

The intuitive answer to this question, I take it, is the following:

The pragmatic system first checks utterances for relevance; if this check is passed, then Quantity-based reasoning can be applied.¹⁸ ‘Jeff Bezos hates Bill Gates’ is not relevant to the QUD; therefore, it is not subjected to Quantity-based reasoning. ‘Mary’s parents gave me conflicting reports,’ by contrast, is relevant; thus, it is subjected to Quantity-based reasoning, which results in the utterance being understood as expressing ignorance.

This answer, however, undercuts the proposal under consideration; indeed, according to this answer, ‘Mary’s parents gave me conflicting reports’ must be recognised as relevant *prior* to any associated ignorance inference being computed. But the moment one commits to this, one is left with nothing—back at square one: in what sense can ‘Mary’s parents gave me conflicting reports’, but not ‘Jeff Bezos hates Bill Gates’, be said to be relevant to the QUD induced by (15)A?

To sum up, if one wishes to maintain that ‘Mary’s parents gave me conflicting reports’ (or ‘Mary’s father told me she’s coming, but Mary’s mother said the opposite’) is not perceived as irrelevant *because* it is understood as expressing ignorance, then one must explain why ‘Jeff Bezos hates Bill Gates’ is not understood as expressing ignorance—and is thus perceived as irrelevant. The explanation attempted above is clearly inadequate, as it presupposes exactly what is at issue. Whether an alternative, non-question-begging explanation can be formulated is an open matter.

As anticipated, there is also an empirical issue. It seems possible to generate cases in which the uttered sentence has no impact on the probabilities of the QUD cells, does not trigger an ignorance inference, and yet is perceived as relevant—a result that calls into question the proposal under consideration. To generate such cases, Fox’s (2014) method for suspending the Maxim of Quantity—which draws on an observation by Grice (1989)—comes in handy. This method consists in creating scenarios in which the speaker is not expected to abide by the Maxim of Quantity.

Consider, for instance, a TV game show in which the audience is presented with the question ‘Is there a ball inside the box?’ and instructed to vote via their phones between two options: ‘Yes, there is a ball inside the box’ and ‘No, there isn’t a ball inside the box.’ As the game unfolds, the host of the show—who is there to build suspense, not to reveal the answer—says:

(16) Is there a ball inside the box? Some people think there is, while some people think there isn’t. Keep voting—we’ll open the box at 5 p.m.!

¹⁸ See, for example, Potts (2006: 7), who explicitly argues that the Gricean maxims come into play one after the other—in particular, that Relevance comes after Quality but before Quantity.

It is clear that, as interpreters, we do not infer—after processing ‘Some people think there is, while some people think there isn’t’ in (16)—that the host doesn’t know whether there is a ball or not inside the box. Yet that sentence seems relevant to the question ‘Is there a ball inside the box?’ This contrast with the case below, which is clearly marked:

(17) # ‘Is there a ball inside the box?’ Jeff Bezos hates Bill Gates. Keep voting—we’ll open the box at 5 p.m.!

Thus, insofar as one thinks that the contrast between (16) and (17) should be explained in the same terms as our original contrast—repeated below—the view that sentences like (18)B₁ are not perceived as relevance violations *because* they convey ignorance becomes difficult to maintain.

(18) A: Is Mary coming to the party tonight?

B₁: Mary’s parents gave me conflicting reports.

[Relevant ✓]

B₂: Jeff Bezos hates Bill Gates.

[Relevant ✗]

3.4 Subject matters

Some philosophers have proposed that sentences, in addition to their regular logical content, also possess their own subject matter or topic (see, for example, Yablo 2014; Fine 2016; Hawke 2016, 2018; Berto 2018; Beddor and Goldstein 2023; Hoek 2025). On this view, the total meaning of a sentence φ —assuming one identifies its regular logical content with its truth-conditions—can be represented as the pair $\langle \varphi^+, \varphi^s \rangle$, where φ^+ is the set of worlds where φ is true ($\varphi^+ = \llbracket \varphi \rrbracket$) and φ^s its subject matter.

Can the notion of subject matter help us make sense of relevance judgments? In what follows, I’ll explore this possibility.

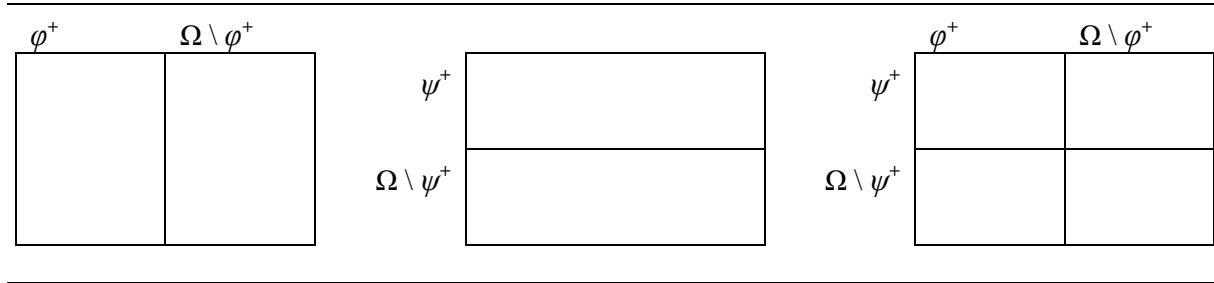
But first things first: what are subject matters? The approach most often taken in recent philosophical work, which has its roots in Lewis (1988), identifies the subject matter of an atomic sentence α with the partition induced by the polar question *Is it true that α ?* The subject matter of a conjunction or disjunction is then derived from the subject matters of its constituent clauses. Following Beddor and Goldstein (2023), Hoek (2025), and others, I take the subject matter of such sentences to be the common refinement of the subject matters of their respective clauses. As for negation, I take it to be subject-matter preserving, as in Perry (2010), Hawke (2018), Beddor and Goldstein (2023), Hoek (2025), and others. The following definition, adapted from Beddor and Goldstein (2023), captures these observations:

(D-IV) Subject Matters

Let $[\cdot]$ be a function mapping sentences to their subject matters; let α range over logically simple (atomic) sentences; and let φ and ψ range over formulas of arbitrary complexity.

- (i) Atomic sentences
 $[\alpha] = \alpha^s = \{\alpha^+, \Omega \setminus \alpha^+\}$
- (ii) Conjunction and disjunction
 $[\varphi \wedge \psi] = [\varphi \vee \psi] = \{i \cap j \mid i \in [\varphi], j \in [\psi], i \cap j \neq \emptyset\}$ ¹⁹
- (iii) Negation
 $[\neg\varphi] = [\varphi]$

Figure 1. Subject matters of: $\varphi, \neg\varphi$ (left); $\psi, \neg\psi$ (centre); and $\varphi \wedge \psi, \varphi \vee \psi$ (right)²⁰



With (D-IV) in place, the following intuitive notion can be formulated:

(D-V) Topical congruence (v. 1)

A sentence φ is *topically congruent* with a QUD Q iff $[\varphi] = Q$.²¹

If one were to present (D-V) as an account of relevance, would it fare better than *PQR*? In particular, can (D-V) help explain why (19)B₁ and (19)B₂, but not (19)B₃, are perceived as relevant under (19)A?

(19) A: Is Mary coming to the party tonight?

- B₁: Mary's father told me she's coming, but Mary's mother said the opposite. [Relevant ✓]
- B₂: Mary's parents gave me conflicting reports. [Relevant ✓]
- B₃: Jeff Bezos hates Bill Gates. [Relevant ✗]

¹⁹ In words: the collection of all non-empty intersections of a partition cell from $[\varphi]$ with a partition cell from $[\psi]$.

²⁰ Adapted from Beddor and Goldstein (2023).

²¹ Drawing on the parthood relation for subject matters (Lewis 1988), a weaker version of topical congruence can also be formulated: a sentence φ is topically congruent with Q if and only if $[\varphi]$ is part of Q . The parthood relation is defined as follows. Let R and Q be two partitions of Ω . Then R is part of Q if and only if every cell of R is a union of cells of Q . For our purposes, the identity-based and parthood-based definitions of topical congruence are equivalent, i.e. they agree in all the cases discussed here.

The answer is ‘no’: none of the answers in (19) are topically congruent with the QUD induced by (19)A. That said, there is something worth noting: (19)B₁ and (19)B₂, unlike (19)B₃, may be argued to contain clauses that, had they been sentences in their own right, would have been topically congruent with the QUD. Thus, to use the notion of subject matter to our advantage, something more powerful than (D-V) appears to be needed—namely, a conception of subject matter that applies not only to full sentences but also to embedded clauses. I won’t attempt to formulate this broader conception of subject matter here, though I take it to be both possible in principle and compatible with the subject-matter framework. Such a conception, once formulated, would make it possible to define the following (weaker) notion of topical congruence:

(D-VI) Topical congruence (v. 2)

A sentence ϕ is *topically congruent* with a QUD Q iff ϕ has a constituent clause k (possibly ϕ itself) such that $[k] = Q$.

(D-VI) helps make sense of the contrast between (19)B₁₋₂ and (19)B₃, which, as discussed, (D-V) cannot handle. Indeed, according to this new definition, (19)B₁ and (19)B₂—at least under certain assumptions—are topically congruent with the QUD induced by (19)A, whereas (19)B₃ isn’t. These assumptions are that, at LF, (19)B₁ is represented as ‘Mary’s father told me she’s coming [*to the party tonight*], but Mary’s mother...’, whereas (19)B₂ is represented as ‘Mary’s parents gave me conflicting reports [*about whether she’s coming to the party tonight*]’. Given these LFs, (D-VI) yields the desired result: by this definition, (19)B₁ and (19)B₂ are topically congruent with the QUD induced by (19)A (both contain the clause ‘she’s coming to the party tonight’), whereas (19)B₃ is not.

(D-VI), it should be noted, is bound to disappoint if presented as a total theory of relevance—indeed, it cannot explain basic cases that *PQR* handles well. Consider, for example (20) and (21) below.

(20) A: Is Mary coming to the party tonight?
 B: She has a difficult exam tomorrow.

(21) A: Is Mary driving to the party?
 B: The roads are piled high with snow.

(20)B and (21)B are relevant to the question asked, despite the fact that neither contains a clause whose subject matter coincides with the QUD. From the perspective of *PQR*, however, there is nothing mysterious here: given a reasonable prior, these answers are expected to affect the probabilities of the QUD cells.

This leaves us in a rather peculiar position: one would have to claim that a sentence is perceived as relevant to a QUD Q if and only if it either (i) shifts the probabilities assigned to the cells of Q (i.e. is

relevant in the *PQR* sense) or (ii) is topically congruent with *Q* (as defined in (D-VI)).²² Could this ‘disjunctive’ approach to relevance be on the right track?

I don’t think so. To begin with, the solution that (D-VI) provides is not general enough; although it makes sense of the contrast between (19)*B*_{1,2} and (19)*B*₃, it is possible to generate analogous contrasts—i.e. contrasts which pose the same challenge to *PQR* as the original cases—using sentences none of whose clauses’ subject matters are identical to the QUD. For example:

(22) A: Is Mary coming to the party tonight?

- | | |
|--|--------------|
| B ₁ : Her father told me she’s in a party mood, but her mother told me she isn’t. | [Relevant ✓] |
| B ₂ : Her parents can’t agree on whether she is in a party mood tonight. | [Relevant ✓] |
| B ₃ : Jeff Bezos hates Bill Gates. | [Relevant ✗] |

(D-VI), quite clearly, cannot be invoked to account for such contrasts.

Moreover, the disjunctive approach just outlined has an overgeneration problem: there are sentences that are not perceived as relevant yet are diagnosed as topically congruent by (D-VI). For example, consider (23):

(23) A: Is Mary coming to the party tonight?

- B: ?? On the assumption Mary is not coming to the party tonight, setting off fireworks may not be a bad idea.

Indeed, (23)*B* does contain the clause ‘Mary is not coming to the party tonight’, whose subject matter is identical to the QUD induced by (23)*A*. Thus, if being diagnosed as topically congruent by (D-VI) were sufficient to secure relevance, then this sentence would be perceived as relevant—and, as far as I can tell, it is not.

Given these considerations, it is not clear that a theory of relevance benefits from incorporating the semantic notion of subject matter—at least, the empirical challenge we have raised against *PQR* does not seem to go away by doing so.

²² Thanks to Dan Hoek and Daniel Rothschild for helpful discussions on this issue.

References

- Bar-Lev, Moshe E. 2024. “Obligatory Implicatures and the Relevance of Contradictions.” *Journal of Semantics* 41 (2): 149–73. <https://doi.org/10.1093/jos/ffae009>.
- Beddor, Bob, and Simon Goldstein. 2023. “A Question-Sensitive Theory of Intention.” *The Philosophical Quarterly* 73 (2): 346–78. <https://doi.org/10.1093/pq/pqac031>.
- Berto, Francesco. 2018. “Aboutness in Imagination.” *Philosophical Studies* 175 (8): 1871–86. <https://doi.org/10.1007/s11098-017-0937-y>.
- Bratman, Michael. 1987. *Intention, Plans, and Practical Reason*. Harvard University Press.
- Büring, Daniel. 2003. “On D-Trees, Beans, And B-Accents.” *Linguistics and Philosophy* 26 (5): 511–45. <https://doi.org/10.1023/A:1025887707652>.
- Carnap, Rudolf. 1950. *Logical Foundations of Probability*. University of Chicago Press.
- Feinmann, Diego. 2020. “Lexical Illusions, (Non-)Maximality, and Invisible Gaps.” PhD Thesis, University of Sheffield.
- Fine, Kit. 2016. “Angelic Content.” *Journal of Philosophical Logic* 45 (2): 199–226. <https://doi.org/10.1007/s10992-015-9371-9>.
- Fox, Danny. 2014. “Cancelling the Maxim of Quantity: Another Challenge for a Gricean Theory of Scalar Implicatures.” *Semantics and Pragmatics* 7 (April): 5:1-20. <https://doi.org/10.3765/sp.7.5>.
- Grice, Herbert Paul. 1975. “Logic and Conversation.” In *Syntax and Semantics: Speech Acts*, Peter Cole and Jerry L. Morgan, vol. 3. Academic Press.
- Grice, Herbert Paul. 1989. *Studies in the Way of Words*. Harvard University Press.
- Groenendijk, Jeroen Antonius Gerardus, and Martin Johan Bastiaan Stokhof. 1984. “Studies on the Semantics of Questions and the Pragmatics of Answers.” PhD Thesis, University of Amsterdam.
- Groenendijk, Jeroen Antonius Gerardus, and Martin Johan Bastiaan Stokhof. 1990. “Partitioning Logical Space.” Annotated handout. University of Amsterdam.
- Hawke, Peter. 2016. “Questions, Topics and Restricted Closure.” *Philosophical Studies* 173 (10): 2759–84. <https://doi.org/10.1007/s11098-016-0632-4>.
- Hawke, Peter. 2018. “Theories of Aboutness.” *Australasian Journal of Philosophy* 96 (4): 697–723. <https://doi.org/10.1080/00048402.2017.1388826>.
- Hoek, Daniel. 2025. “Minimal Rationality and the Web of Questions.” In *Unstructured Content*, edited by Peter van Elswyk, Dirk Kindermann, Cameron Domenico Kirk-Giannini, and Andy Egan. Oxford University Press.
- Keynes, John Maynard. 1921. *Treatise on Probability*. Macmillan & Co.
- Lewis, David. 1988. “Statements Partly about Observation.” *Philosophical Papers* 17 (1): 1–31. world. <https://doi.org/10.1080/05568648809506282>.
- Morris, Stephen. 1995. “The Common Prior Assumption in Economic Theory.” *Economics and Philosophy* 11: 227–53.
- Perry, John. 2010. “Possible Worlds and Subject Matter. Discussion of Barbara H. Partee’s Paper ‘Possible Worlds in Model-Theoretic Semantics: A Linguistic Perspective.’” In *Possible Worlds in Humanities, Arts and Sciences*, edited by Allén Sture. De Gruyter. <https://www.degruyterbrill.com/document/doi/10.1515/9783110866858.124/html?lang=en>.
- Potts, Christopher. 2006. “How Far Can Pragmatic Mechanisms Take Us?” *Theoretical Linguistics* 32 (3): 307–20. <https://doi.org/10.1515/TL.2006.019>.
- Roberts, Craige. (1996) 2012. “Information Structure in Discourse: Towards an Integrated Formal Theory of Pragmatics.” *Semantics and Pragmatics* 5. <https://doi.org/10.3765/sp.5.6>.
- Roberts, Craige. 2012. “Information Structure: Afterword.” *Semantics and Pragmatics* 5 (December): 7:1-19. <https://doi.org/10.3765/sp.5.7>.
- Russell, Benjamin. 2012. “Probabilistic Reasoning and the Computation of Scalar Implicatures.” Brown University.
- Simons, Mandy, Judith Tonhauser, David Beaver, and Craige Roberts. 2010. “What Projects and Why.” *Semantics and Linguistic Theory* 20 (0): 309–27. <https://doi.org/10.3765/salt.v20i0.2584>.

- Stalnaker, Robert. 1974. “Pragmatic Presuppositions.” In *Semantics and Philosophy*, edited by Munitz Milton K. and Peter Unger. New York: New York University Press.
- Stalnaker, Robert. 1978. “Assertion.” In *Syntax and Semantics (New York Academic Press)*, vol. 9. New York: New York Academic Press.
- Thomason, Richmond H. 1990. “Accommodation, Meaning, and Implicature: Interdisciplinary Foundations for Pragmatics.” In *Intentions in Communication*, edited by Philip R. Cohen, Jerry Morgan, and Martha E. Pollack.
- Yablo, Stephen. 2014. *Aboutness*. Princeton University Press.